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Multi-Criteria Decision Analysis for the Evaluation of Nuclear Reactor Designs

Tesi di Laurea Magistrale in
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Author: **Simone Pagliuca**

Student ID: 247812
Advisor: Prof. Stefano Lorenzi
Co-advisors: Prof. Russell McKenna, Dr. Peter Burgherr, Dr. River Huang

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Abstract

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Contents

Introduction

A brief History of Nuclear Power

Origins

The 1930s and 1940s witnessed rapid advancement in nuclear science. In 1932 James Chadwick discovered of the neutron [14] and a year later Leo Szilard hypothesized fission chain reactions. In 1938, Otto Hahn and Fritz Strassmann produced barium by bombarding uranium with neutrons, demonstrating nuclear fission [34]. This breakthroughs culminated in Enrico Fermi's team achieving the first controlled nuclear chain reaction in the Chicago Pile-1 in 1942.

Initial advancements were speed up by World War II though the manhattan project and lead to the atomic bombings of Hiroshima (6 August 1945) and Nagasaki (9 August 1945) but the recognition of the paeceful potential of nuclear technology lead to the famous speech "Atoms for Peace" by USA President Eisenhower in 1953 to the United Nations [18] which established the foundation for international nuclear cooperation. The International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM) were established 4 years after, in 1957, and were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

In the 1950s numerous reactor designs were born driven by both, civilian and military necessities. The Experimental Breeder Reactor I (EBR-I) in Idaho (USA) first produced electricity in 1951 (powering four light bulbs), followed by the Obninsk Nuclear Power Plant (USSR) in 1954, which became the first to feed approximately 5 MWe into a local power grid. The world's first commercial-scale nuclear power station, Calder Hall-1 (UK), began operation in 1956, connected to the national grid with a capacity of 50 MWe. The United States pioneered the light water reactor development (LWR), originally developed in the military for naval propulsion, launching the USS Nautilus in 1954. The first civilian application of the LWR technology in the USA was the Shippingport Atomic Power Station, which started operation in 1957 with a capacity of 60 MWe [40].

To garantee their independence, each country developed their own nuclear reactor, which resulted resulted in technological diversification. The United Kingdom developed gas-cooled Magnox reactors, optimized for both electricity and plutonium production, while the Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design. Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the 1962 Rolphton demonstration reactor and the first commercial CANDU reactor at Douglas Point in 1968 [40]. Regarding LWRs, the USA developed the BWR (Boiling Water Reactor) and adapted their naval reactors to create commercial PWRs (Pressurized Water Reactor), at the same time the USSR developed their own version of the same technology, the VVER ("Water-Water Energetic Reactor"), with some design differences that still persist to this day.

Between the 60s and 70s a numerous countries commissioned their first nuclear reactors: Belgium (1962), Italy (1963), Japan (1963), Sweden (1964), Switzerland (1969), Slovakia (1972), Argentina (1974), Bulgaria (1974) and Armenia (1976) [40]. Following the 1973 oil crisis, the deployment of nuclear power plants was accelerated in some countries as an energy security strategy. Most notably France implemented the Messmer Plan, named after the Prime Minister at the time, to greatly expand the nuclear plant fleet [29].

Two setbacks: 1986 and 2011

Nuclear expansion continued through the 1980s until 1986. As news from the Chernobyl disaster came to western countries, fear of the potential risks of nuclear power intensified. The public opposition grew significantly so that, even if safety and regulatory updates were implemented, several countries decided to slow down, reduce or halt althogether their nuclear programs, such was the case of Italy after the results of the 1987 referendum, which resulted in the complete phase-out of nuclear power [50, 40].

By the 1990s, multiple western countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected 10 reactors between 1990-1999, while the United States and United Kingdom only deployed 4 and 1 reactors respectively during the same period. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since continued expansion of its reactor fleet [40].

In the early 2000s a nuclear renaissance, driven by concerns over climate change, energy security, and rising fossil fuel prices, lead to the design of Generation III+ reactors, such as the European Pressurized Reactor (EPR) or the US AP-1000. However after the 2011

Fukushima Daiichi accident Western nations expirenced a significant setback: Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [49]. In the following years projects faced substantial delays, cost overruns, and public opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case: experiencing budget increases to more then four times original estimates and construction timelines tripled from initial projections [53].

Contrastingly, Eastern nations pursued an opposite trajectory. China, having initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991), has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within the coming decade [40].

Recent years

The current political scenario

In 2020, the COVID-19 pandemic exposed vulnerabilities in global economic systems and Russia's invasion of Ukraine in 2022 further revealed Europe's energy dependencies [10, 78, 9]. At the same time, decarbonization goals to combat climate change have prompted Western nations to reconsider nuclear energy as a valid power source able to provide low-carbon, reliable power while enhancing energy independence and system resilience [22].

This strategic reassessment of nuclear is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [23], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [84], are a clear indicator of the renewed interest in nuclear technologies.

This historical overview demonstrates that the initial decision to adopt nuclear power is a strategic and political choice, deeply influenced by historical context. However, the subsequent selection of a specific reactor design involves complex trade-offs across technical, economic, regulatory, social, environmental, and geopolitical aspects. Moreover the choice is inevitably tied to specific national energy requirements.

The state of Nuclear Designs in 2025

As mentioned before, several techologies have been developed thoughout the years, with some being split between opposing political sides, most notably the western implementation of the PWR technology and its soviet counterpart, the VVER, which is nowadays the most common reactor type by far, counting 306 out of the total 416 reactors worldwide. Followed by heavy water reactors (46) and BWRs (43). The current generation is usually reffered to as Generation III+ reactors, variations of previous iterations from the 90s (Gen III) updated to the latest safety standards. Generation IV reactors are mostly under development or at demonstation phases and employ innovative means of colling, like molten salts or molen metals. Small Modular Reactors (SMRs) on the other hand represent a different approach to the building strategy of the power plant itself. Since the beginning, the nuclear industry has exploited economies of scale by building larger and larger reactors. Unfortunatley in the last years, nuclear commissioning has reach the status of "Mega Projects". These term is used to refer to very large projects where economies of scale tend to fail due to the complexity of the task at hand, resulting in massive delays and cost overruns [27, 11]. To counter this phenomena the industry is trying to develop SMRs: smaller, standardized reactors that can be built mostly off site to take advantage of economies of multiples. Some SMR designs are based on novelty generation IV techonlogies while others take advantage of well known water-based designs.

Challenges and solutions of our days

The present landscape presents policymakers with significant decision-making challenges. Renewed interest in nuclear power, geopolitical uncertainties and technological innovations, complicates the selection of an optimal reactor design. While nuclear commitments involve decade-long development timelines, 60-year operational lifespans, and substantial financial investments, modern energy policies must also consider evaluation of electricity demand, public dynamics, and supply chain vulnerabilities.

To address this complex decision-making environment, analytical frameworks have been developed, notably Multi-Criteria Decision Analysis (MCDA) which provides systematic support for evaluating alternatives when multiple, conflicting, criteria must be considered.

1 | Literature Review and Objectives

1.1. Literature Review

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge:

Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [16] [7]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [1]. Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without sistematic comparison [17].

A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [43], while Locatelli and Mancini (2011) compares large and small reactor designs [46]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models (NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [8]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [38].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [35, 67, 5]. In contrast, Castelazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT) to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [74].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [94]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [82]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [33].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack considerration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

This study addresses these research gaps by applying Multi-Criteria Decision Analysis (MCDA), specifically employing Multi-Attribute Value Theory (MAVT), to compare reactor designs. The analysis explicitly incorporates country-specific conditions and examines their influence on the ranking of different nuclear installations.

1.2. Applying MAVT to choose a Reactor

This study evaluates a selected group of nuclear reactor designs to determine their suitability for different national contexts. By comparing four European countries with distinct energy profiles, the analysis will also explore how factors such as demand patterns, grid flexibility, and policy priorities, shape the ranking of reactor designs.

The methodology relies on Multi-Criteria Decision Analysis (MCDA) to systematically evaluate reactor designs by taking into account real world data, estimates, and expert judgment. Specifically, this study uses Multi-Attribute Value Theory (MAVT) which, first presented by Keeney in 1996 [42], transforms qualitative and quantitative data into comparable metrics (values) by means of value functions, then aggregates values by hierarchical weighting to produce a composite score for each option.

Here i will put the signposts to do the overview of the thesis structure... once i wrote it all... In the following section we are going to discuss the selection of reactor models then.... etc etc etc

2 | Preparation

2.1. Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [58], which catalogs operational, under-construction, and decommissioned reactors globally.

The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.

Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.

From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [40]. The selection process and its outcomes are summarized in Table 2.1.

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [2], was consulted to identify advanced reactor models classified as under design, under regulatory review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.2.

The overall selection produced a set of six reactors, three Generation III+ and three SMRs. Their main characteristics are summarized in Table 2.3. The most notable differences are the integrated designs of the BWRX-300 and Nuscale SMR which eliminate the external primary loop in favour of steam generator inside the reactor pressure vessel. This allows for compact construction and allows to exploit natural circulation for cooling. Then the Nuscale SMR is the only model designed to be deployed in a multi unit configuration, with up to 12 modules in the same containment building. Maybe here add a paragraph for each design to explain their main features.

Reactor Name	Country	Technology-sharing	Commercial Operation	Up to Date	Decision
APR1400	S.Korea	✓	✓	✓	**Selected**
CAREM Prototyp	Argentina	✓	✗	✓	Rejected
PRE KONVOI	Germany	✓	✓	✗	Rejected
EPR 1750	France	✓	✓	✓	**Selected**
AP1000	USA	✓	✓	✓	**Selected**
ABWR	Japan	✓	✗	✓	Rejected
APWR	Japan	✓	✗	✓	Rejected
EPR	France	✓	✓	✗	Rejected
OPR-1000	S.Korea	✓	✓	✗	Rejected

Table 2.1: Selection of reactor models for analysis

Reactor Name	Country	Used	Reasoning
APR+	S.Korea	✗	As reported in the manufacturer website [21] this design is mean to be an "indigenous reactor cleared of obstacles to export"
ATMEA	Japan, France	✗	Joint venture was abandoned in 2019 [6]
BWRX300	USA, Canada	✓	Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [55] [56].
i-SMR	S.Korea	✗	Never expressed interest in development outside south Asia and the middle east, no updates since early 2024.
iMSR400	US	✗	No interest overseas, project seems to be at an early stage of development.
Natrium	US	✗	Demo plant [81].
Nuward	France	✗	After the change in direction of 2024 the project has been set back [6].
Nuscale	US	✓	Even if there was some financial controversy in 2024 [6]. the project seems to be still moving forward in terms of goverment founding [6] and regulatory review [6].
PRISM	US	✗	GE-Hitachi considers it a "concept ... currently being put into practice in two reactors: Natrium and ARC-100" [6].
Rolls Royce SMR	UK	✓	It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[6].
SMART	S.Korea, Saudi Arabia	✗	Plan to export only to south-east asia and middle east [57].

Table 2.2: Selection of advanced reactor models for analysis

Reactor Name	Type	Construction	P_e (MWe)	Supplier
APR1400	PWR	1 Unit	1340	KEPCO (S.Korea)
EPR 1750	PWR	1 Unit	1630	Framatome (France)
AP1000	PWR	1 Unit	1120	Westinghouse (USA)
BWRX300	i-BWR	1 Unit	300	GE-Hitachi (USA-Canada)
Rolls Royce SMR	PWR	4 to 12 Units	470	Rolls Royce (UK)
Nuscale	i-PWR	1 Unit	77	Nuscale Power (USA)

Table 2.3: Final selection of reactor models for analysis

2.2. Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[6]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[6], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [6], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [6], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

2.3. Attribute Definition

Throughout this section we will define the indicators.

2.3.1. Technical parameters

...

2.3.2. MOX Fuel Compatibility

Mixed Oxide (MOX) fuel is produced through spent nuclear fuel reprocessing and offers a solution for partially closing the nuclear fuel cycle in thermal reactors. While this process generates some secondary waste, it provides several strategic advantages: reducing final nuclear waste volumes, minimizing mining waste, and decreasing dependence on uranium imports. The latter is particularly relevant for Europe, which lacks significant domestic uranium deposits. However, reprocessing also presents proliferation risks since the extracted plutonium could potentially be diverted for weapons production. Due to these competing factors, nuclear fuel reprocessing is a fundamental strategic decision; the United States has foregone this practice, while France and UK have implemented the process in their nuclear programs [15].

Given the European context of this study, MOX compatibility is considered a relevant reactor attribute. The evaluation methodology assigned each reactor design an initial score (0-10) based on its maximum MOX core fraction capability, followed by qualitative adjustments reflecting design-specific features. The complete scoring process and results are detailed in Table 2.4.

Reactor Model	**MOX[%]**	**Considerations**	**Score**	Sources
APR1400	33	-	3/10	[2]
EPR 1750	100	50% without modifications, to reach 100% "design adaptations which are not significantly costly" are required.	9/10	[2, 4]
AP1000	50	50% without modification, to reach 100% major modifications are required	6/10	[2, 26]
BWRX-300	0	GNF-2 Not manufactured with MOX	0/10	[12]
Rolls Royce SMR	0	Rolls Royce states that they are not planning to use MOX	0/10	[70]
NuScale	0	Not licensing for MOX but a study reveals support is possible without design modifications	1/10	[89, 61]

Table 2.4: MOX fuel compatibility scores and maximum MOX fraction

2.3.3. Technological Readiness

Expert consultation

2.3.4. Safety

To be discussed

2.3.5. Capital Costs

DRAFT

Option	**PROs**	**CONs**
Use NOAK estimates	Use of estimates in all cases, comparable values	It's not realistic and not fair towards more mature designs
Use FOAK estimates/data	Better data for three large reactors	We are mixing up real data and estimates
Use NOAK and apply FOAK-to-NOAK conversion	Comparable values, with awareness of current know-how	rely on a model
Present NOAK and FOAK to experts along with technological readiness overview and let them evaluate	As before	rely on expert judgment

Table 2.5: Options overview

The same applies to commissioning time.

It is important to note that these costs refer to an Nth of a kind installation and will not be representative of the FOAK costs. Moreover as detailed by Rothwell 2022 [73] the costs of recent nuclear installations in Europe and the US have been significantly higher than those announced at the start of the project. For this reason we will apply a FOAK-to-NOAK factor to better represent the real costs. For example the EPR-1600 announced costs for Flamanville-3 was 1886 USD_{2020}/kW_e but have risen to 8600 USD_{2020}/kW_e [54]. All Values are summarized in Table 2.6.

For reference, the capital costs of the same plants as the previous section have been reported by the US Energy Information Administration [85].

We want to clarify that the data has all been converted to 2020 USD since the main source of data [77] and [54] reports all costs in this currency and year.

Reactor Model	**Capital Cost of NOAK [$/kWe]**	**Capital Cost of FOAK [$/kWe]**	Data Source
APR-1400	1828	2410	[77, 54]
EPR-1750	2150	8600	[77, 54]
AP-1000	2000	8600	[77, 54]
BWRX-300	2318	3467	[69]
Rolls Royce SMR	6050	7408	[68]
NuScale	2850	3600	[62]
Other Power Plants			
Onshore Wind	-	1265	[85]
Solar PV + Battery	-	2175	[85]
Hydroelectric	-	6007	[85]

Table 2.6: Capital costs for selected reactor models

2.3.6. Commissioning Time

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See observations in the previous section

I would apply a +0.5 years to planned commissioning time on the APR and AP based on the planned commissioning time of the EPR that differs by 0.5 between UK and Chinese plannings.

Reactor Model	**Planned Commissioning Time**	**Last unit Commissioning Time**	Data Source
APR1400	5.0 in Korea	10 and 8 in Korea	[53]
EPR 1750	5.0 in Europe, 4.5 in China	9 in Taishan, 17 in Flamanville and Olkiluoto	[53]
AP1000	5.0 in USA	9 in Sanmen, 10 and 11 at Vogtle	[53]
BWRX-300	$2 \sim 3$	NA	[31]
Rolls Royce SMR	4.0	NA	[71]
NuScale	3.0	NA	[63]

Table 2.7: Technological readiness and planned commissioning time

2.3.7. Operational Costs

Lascks presentation of the attribute

Steigerwald et al. (2023) [77] provides comprehensive cost analyses for future nuclear installations, including capital and operational expenditures for both SMRs and large reactors. The study reports operational costs in USD_{2020}/MWh_e for EPR and APR1400 designs, as well as a "Long-Term US" category which this analysis uses as AP1000 costs. For SMRs, operational costs were originally presented in power-based units (USD_{2020}/MW_e), requiring conversion to energy-based measurements (USD_{2020}/MWh_e) using a Gaussian distribution of capacity factors ($\mu = 90\%$, $\sigma = 5\%$) according to:

$$\frac{USD_{2020}}{MWh_e} = \frac{USD_{2020}}{MW_e} \cdot \frac{1}{CF \cdot 8760 \frac{h}{year}}$$

The same study reports fuel costs of $6.23 USD_{2020}/MWh_e$ for PWR-based technologies with the exceptions of NuScale at $7.15 USD_{2020}/MWh_e$ and the only BWR reactor at $29.55 USD_{2020}/MWh_e$. For comparative context, operational cost estimates for utility-scale onshore wind, hydroelectric, and solar PV (with single axis tracking and DC battery storage) were obtained from the U.S. Energy Information Administration [85]. All cost data have been standardized to 2020 USD to maintain consistency with the primary data sources [77, 54], values are presented in Table 2.8.

Reactor Model	**Operative Cost [$/MWh]**	Data Source
APR-1400	18.44	[77]
EPR-1750	14.26	[77]
AP-1000	18.69	[77]
BWRX-300	18.30 ± 0.82	[77]
Rolls Royce SMR	65.68 ± 2.94	[77]
NuScale	22.76 ± 1.02	[77]
Other Power Plants		
Onshore Wind	28.08	[85]
Solar PV + Battery	33.33	[85]
Hydroelectric	28.49	[85]

Table 2.8: Economic costs for selected reactor models

2.3.8. Potential Impact on the energy market

To write

2.3.9. Potential Impact on greenhouse gas emissions

To write

2.3.10. Waste

To write

2.3.11. Size - Nominal output power

To write

2.3.12. Load Following Capabilities

Load-following capability can be quantitatively assessed using characteristic curves that represent ramp-up and ramp-down rates as a function of load (expressed as percentage of nominal power $\%P_{nom}$), as illustrated in Figure 2.1. To facilitate comparative analysis, these performance curves are reduced to scalar values through integration of the area under said curve.

For benchmarking purposes, load-following capabilities were calculated for two modern gas turbine configurations: A heavy-duty combined cycle plant featuring two GE Vernova 9HA.02 gas turbines (2015 data) [32] and an equivalent plant configuration utilizing aeroderivative GE LM2500XPRESS+G5 DLE UPT gas turbines (2025 data) [30]. Hydroelectric plants are also included as a reference for the highest flexibility, a 1989 publication by the US Office of Technology Assessment [83] reports that the response rate of large (>60MW) hydroelectric plants ranges from 4 to 6 $\%\mathrm{P}_{nom}$/s, which translates to 240 to 360 $\%\mathrm{P}_{nom}$/min. A graph from a 1994 conference reported in a technical analysis from 2009 shows that some hydro plant can achieve 0 to 100% in much less then a minute and from black startup to full power in around 5 minutes For the comparison we indicate greater then 100%. [75, 51]

To establish an upper benchmark hydroelectric plants were included in the comparative analysis. A 1989 U.S. Office of Technology Assessment report [83] documents that large (>60 MW) hydroelectric facilities demonstrate remarkable load-following capabilities, with response rates ranging from 4 to 6 $\%\mathrm{P}_{nom}$/s (equivalent to 240-360 $\%\mathrm{P}_{nom}$/min). Additional performance data from a 1994 conference presentation, cited in a 2009 technical analysis [75, 51], indicates that certain hydroelectric installations can achieve 0-100% load changes in under one minute and black start to full power operation in approximately five minutes. For the purposes of this comparison, hydroelectric flexibility is conservatively represented as >100 %P_{nom}/min.

The results of these analyses are presented in Table 2.9.

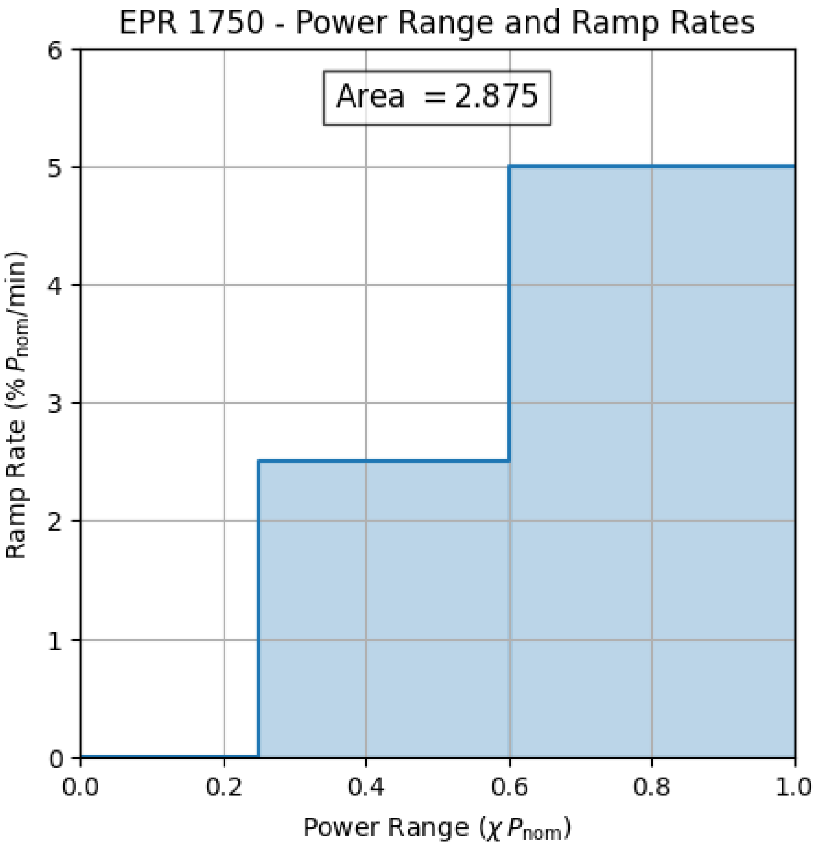

Figure 2.1: Example load following capability curve for EPR-1750

Reactor Model	**Load Following Capability** $\left[\frac{\%P^2_{nom}}{min}\right]$	Data Source
APR1400	0.21	[2]
EPR 1750	2.88	[64]
AP1000	4.25	[2]
BWRX-300	0.25	[2]
Rolls Royce SMR	2.00	[2]
NuScale	2.00	[20]
GE 9HA.02 2xCC	7.02	[32]
GE LM2500XPRESS+G5 2xCC	40.54	[30]
Hydroelectric	>100	[83, 75, 51]

Table 2.9: Load following capabilities and power output

3 | Analysis

4 | Results

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A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, . . .) which supplement the main results contained in the previous chapters.

B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

List of Figures

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List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m

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